SGA 2: Local Cohomology and Lefschetz Theorems

Exposé VIII: The Finiteness Theorem

Bounded source-aligned checkpoint — 19 July 2026

Authority. Corrected French lines 2828–2854; original printed pp. 94–95; physical source-PDF p. 83; recomposed running p. 75. The printed-page-95 marker occurs at French line 2847. Blank line 2855 is excluded; French line 2856 begins the $n > 1$ case and is the exact continuation cursor. The compiled French reader is generated from the same corrected edition and is page/layout evidence, not independent corroboration.

If $n = 1$, assume that $i_*(\mathcal{F}|_U)$ is coherent. We argue by contradiction and suppose that there exists $x \in U$ such that $c(x) = 1$ and

$$\operatorname{prof} \mathcal{F}_x = 0,$$

i.e. $x \in \operatorname{Ass} \mathcal{F}_x$. Let $y \in \overline{\{x\}} \cap Y$ be such that

$$\dim \mathcal{O}_{\overline{\{x\}},y} = 1.$$

Set

$$A = \mathcal{O}_{X,y} \qquad \text{and} \qquad X' = \operatorname{Spec}(A).$$

Perform the base change $v\colon X' \longrightarrow X$, which is flat:

$$\begin{array}{ccc} U' = X' \times_X U & \xrightarrow{v'} & U \\ {\scriptstyle i'}\big\downarrow & & \big\downarrow{\scriptstyle i} \\ X' & \xrightarrow{v} & X. \end{array} \tag{2.7}$$

The morphism i is separated, since it is an immersion, and is of finite type, since it is an open immersion and X is locally Noetherian. Since the base change is flat, EGA III, 1.4.15 gives an isomorphism

$$v^*(i_*(\mathcal{F}|_U)) \approx i'_*(v'^*(\mathcal{F}|_U)). \tag{2.8}$$

French printed p. 95 Denote by $\underline{x}$ (respectively, $\underline{y}$) the ideal of A corresponding to x (respectively, y).

Set $\mathcal{G} = v'^*(\mathcal{F}|_U)$. Then $\mathcal{G}$ is coherent and $\underline{x} \in \operatorname{Ass} \mathcal{G}$, so there exists a monomorphism

$$\mathcal{O}_{\overline{\{x\}}} \longrightarrow \mathcal{G};$$

consequently, $i'_*(\mathcal{O}_{\overline{\{x\}}}|_{U'})$ is coherent. By the choice of y, $\dim(A/\underline{x}) = 1$, and therefore the support of $\mathcal{O}_{\overline{\{x\}}}$ is precisely

$$\overline{\{\underline{x}\}} = \{\underline{x}\} \cup \{\underline{y}\},$$

since $\overline{\{\underline{x}\}} = \operatorname{Spec}(A/\underline{x})$ as a scheme. It follows that

$$(\mathcal{O}_{\overline{\{x\}}}|_{U'})(U') = \operatorname{Frac}(A/\underline{x}),$$

the field of fractions of $A/\underline{x}$, and

$$i'_*(\mathcal{O}_{\overline{\{x\}}}|_{U'})(X') = \operatorname{Frac}(A/\underline{x}).$$

But $\operatorname{Frac}(A/\underline{x})$ is not a finitely generated A-module, because $\underline{x}$ is different from the maximal ideal of A. This is a contradiction.